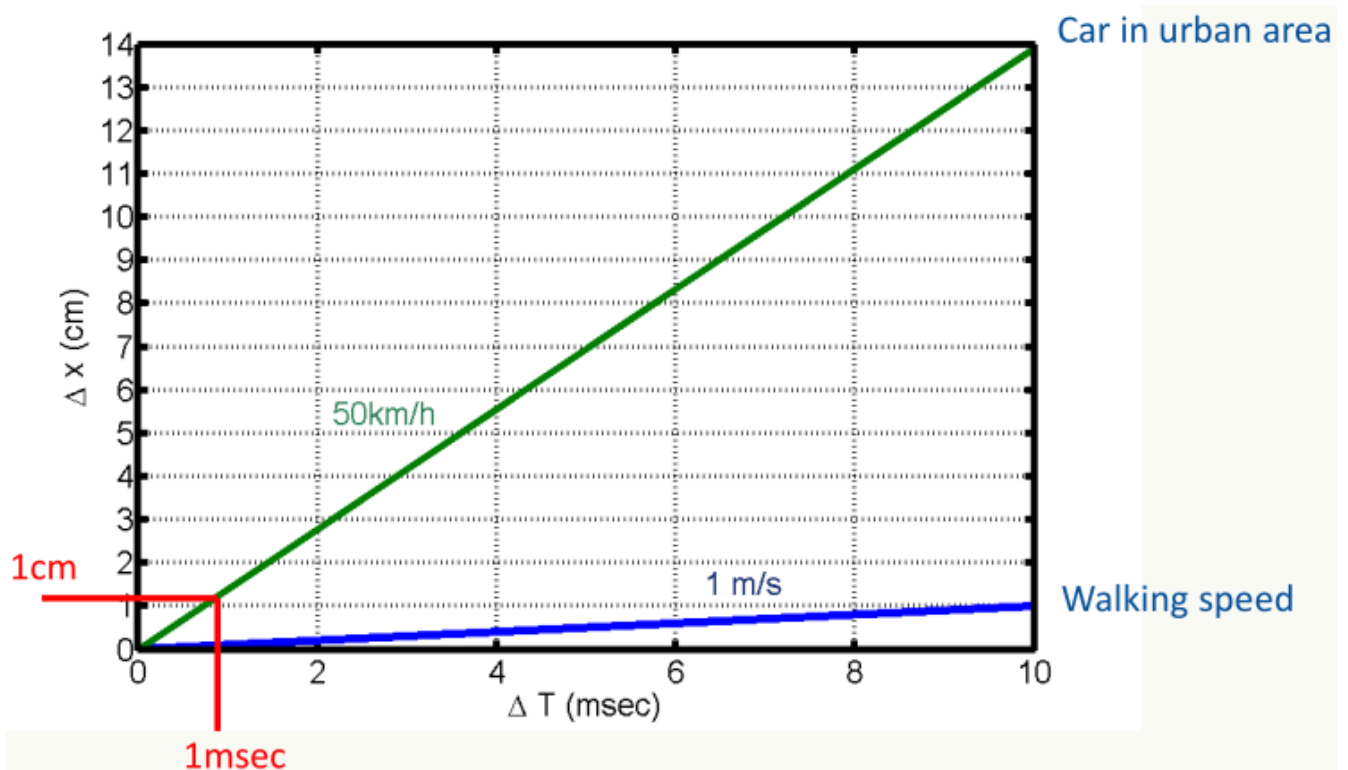


L6: Sensor Synchronisation

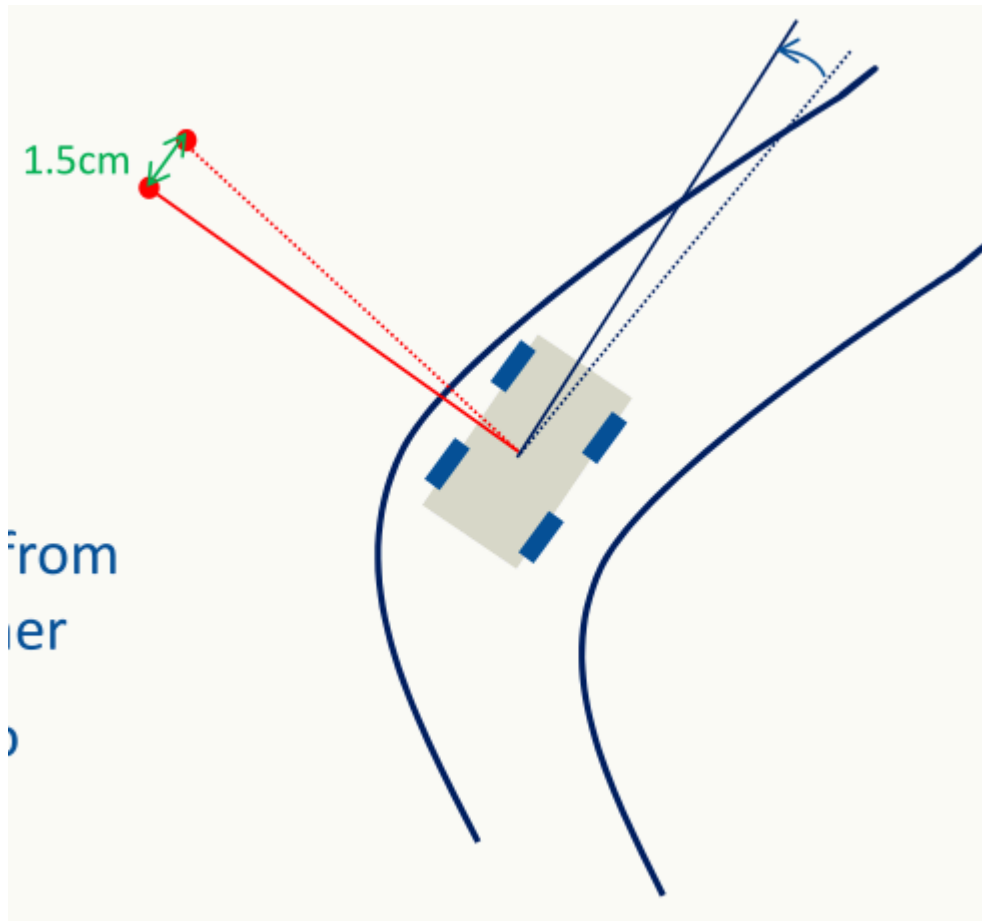
Sensor	Typical rate
GNSS receiver	1-10Hz
IMU	100-10000Hz
Laser Scanner	100kHz-1MHz (point rate) 10-100Hz (profile rate)
Camera	1-100Hz
Depth Cameras	1-100Hz

- Time difference $dt = dt' - dt''$ between
 - Scanner observation
 - Trajectory parameter
 - -> Wrong assignment
 - -> Wrong Transformation
- Error depends on moving speed and rotation rate



- 1msec
- Rotation Error:
 - Assume 90° turn in 1 second
 - rotation rate 90°/sec
 - $dt = 1ms$ leads to 0.1° angle error
 - 10m distance from object to scanner

- 0.1° Angle error leads to 1.5cm position error



- Performing action at specific position needs:
 - High synchronisation accuracy, depending on movement speed

Synchronisation Tasks

1. Stamping
 - Timestamp to a measurement
2. Synchronisation
 - Synchronisation of multiple clock

1. Stamping

1. Option: Stamped data
 - Sensor already provides properly tagged measurements
 - GNSS always UTC stamped
 - IMU, containing GNSS consequently too
2. Option: Receive pulse
 - Sensor provides electronic signal at measurement
 - Higher GNSS receivers provide electronic signal upon measurement

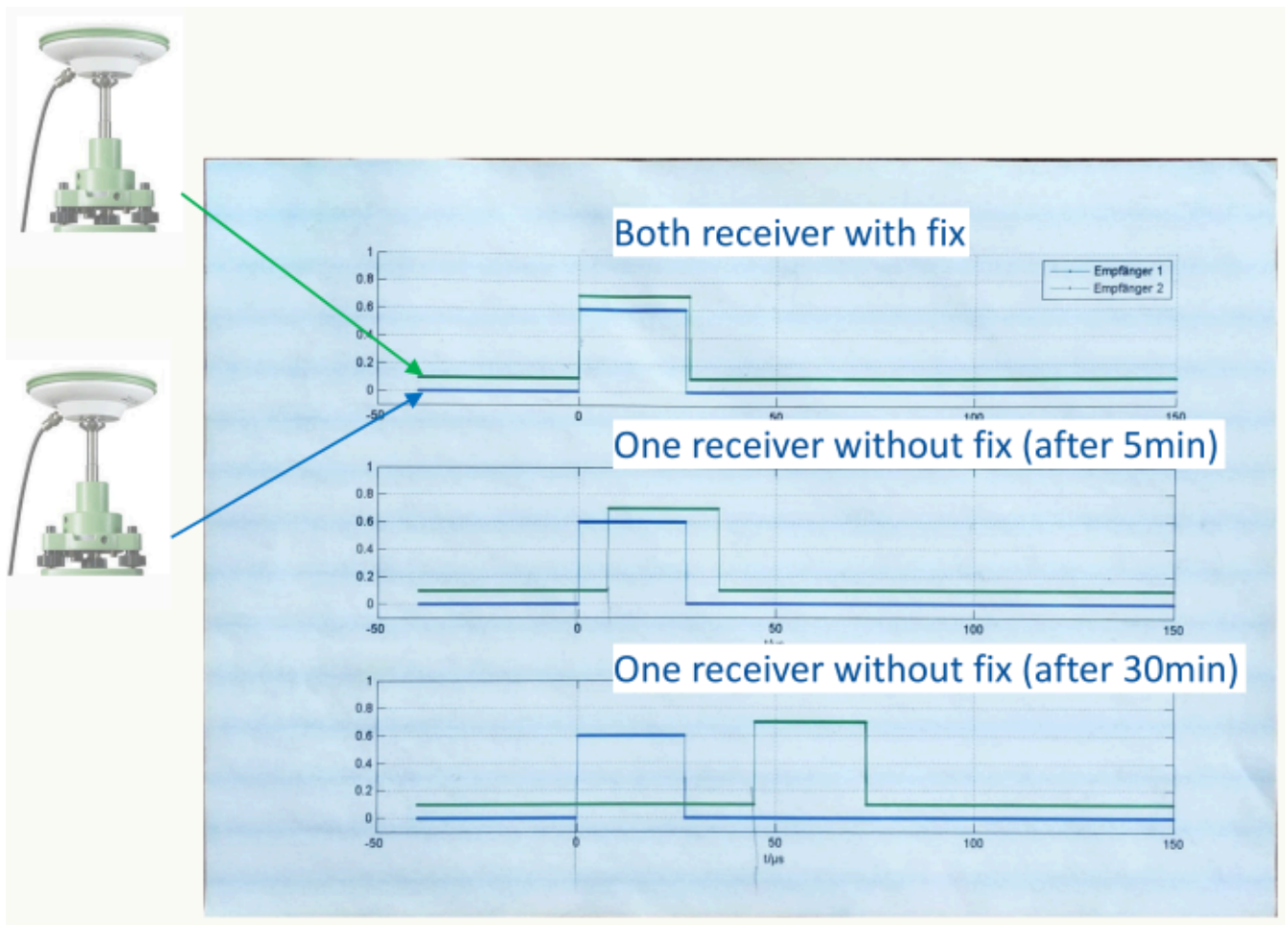
- PPS Pulse per second
 - Usable for synchronising clocks
- Industry cameras provide shutter synchronised trigger
- Some Cameras provide hot shoe flash synchronisation
 - might limit exposure time
- Laser Scanner: Signal when Profile crosses 0°

3. Option: Trigger

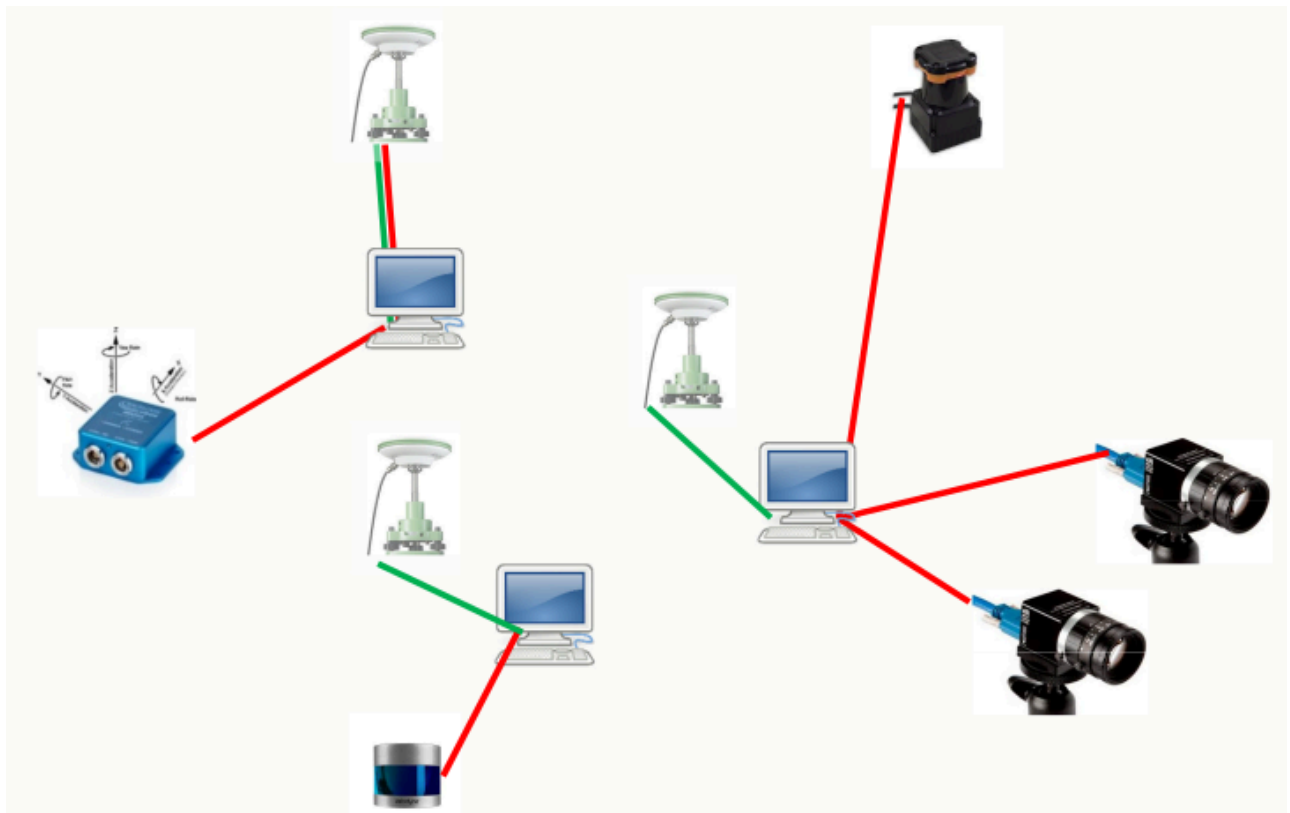
- Sensor receives trigger from electronic signal
- Sensor performs measurement
- Cameras:
 - Industry and some consumers provide trigger input signals
 - Still question $t_{\text{exposure}} - t_{\text{trigger}}$
- Laser Scanners:
 - Scanner mirror rotation sometimes synchronisable
 - Only synchronises oscillations, not single events
- USB connected sensors without extra sync possibilities in general problematic
 - span: 5ms

2. Synchronisation

- UTC: Atomic time with leap seconds (distance <0.9s)
- GNSS:
 - Bias estimated with about 10 Hz
 - Update internal clock with offsets and provides PPS
 - PPS stable within nanoseconds

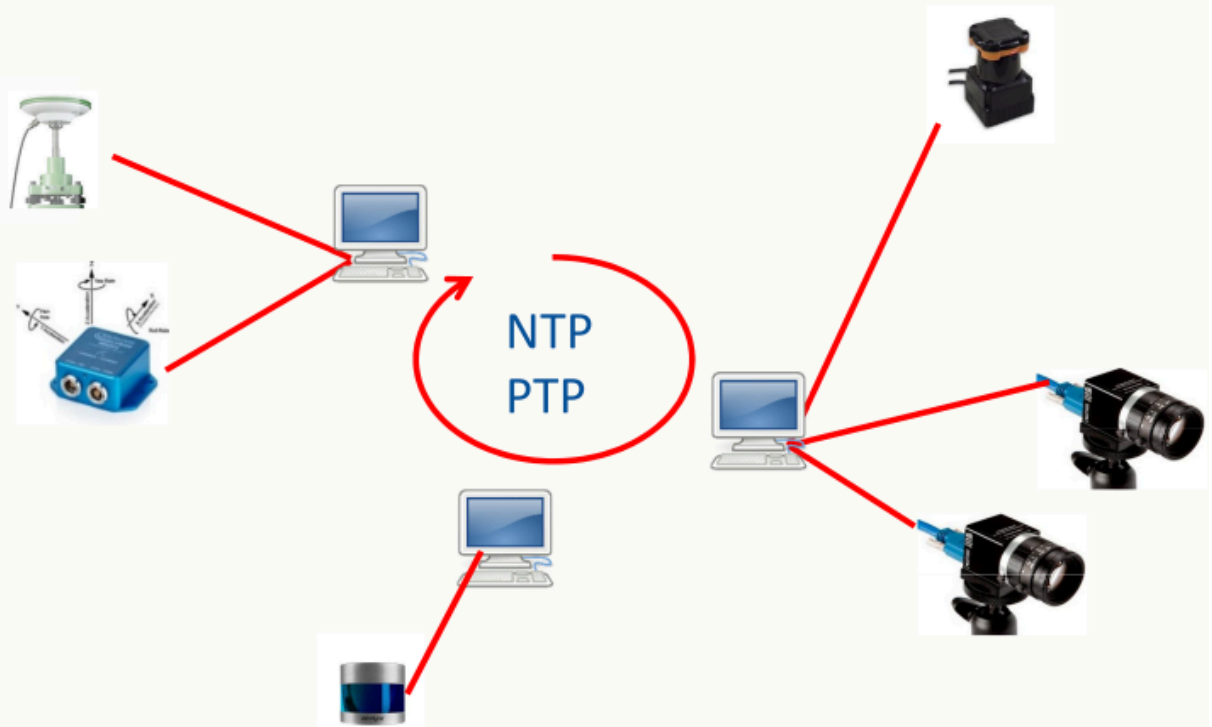


1. Synch all clocks with GNSS



2. Synch clocks via clock sync protocol

OPTION 2 (TIME PROTOCOLS): SYNCHRONIZE CLOCKS VIA CLOCK SYNC PROTOCOL



o

PTP - Precise Time Protocol

- Master and Slave structure
- Synch and Follow-up msg to measure delay
- nano-micro seconds accuracy
 - depends on hardware/software implementation

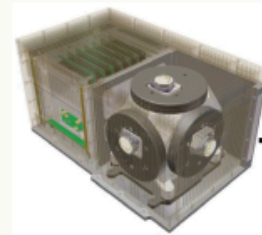
NTP - Network Time Protocol

- Hierarchical structure
- offset calculation between with different algorithms
- Synchronisation is accurate to $10\mu s$
- Convergence time not ideal for sensors



PPS

NMEA (GGA message contains UTC time)



$\sim 10^3$ kHz

Data file
timestamp (UTC) x y z
timestamp (UTC) x y z
timestamp (UTC) x y z
...

~ 1 kHz

Data file
timestamp (UTC) acc_x acc_y acc_z ...
timestamp (UTC) acc_x acc_y acc_z ...
timestamp (UTC) acc_x acc_y acc_z ...
...

Interpolate